

Engineering Change Order ECO-0412

Title	Arm lift position sensor replacement (supplier end-of-life)
ECO number	ECO-0412 · Rev A
Originator	Electrical Engineering, Northfield Body Works
Date raised	2026-02-03
Product line	ASL-200 automated side-loader body
Priority	Production - supplier last-time-buy 2026-06-30

Reason for change

The supplier has issued an end-of-life notice for the arm lift angle sensor P/N 55-1180-0 (Hall-effect rotary, 0-5 V ratiometric). Last-time-buy quantity covers production through serial ASL2-26-04099. The supplier's replacement P/N 55-1180-1 is mechanically identical (same flange, shaft and connector) but has a **0.5-4.5 V** output over the same angular range, with diagnostic bands outside that range.

Affected parts

	Old	New
Part number	55-1180-0	55-1180-1
Output	0-5 V ratiometric	0.5-4.5 V (ratiometric to 5 V supply)
Angle at output limits	-20° @ 0.0 V, 180° @ 5.0 V	-20° @ 0.5 V, 180° @ 4.5 V
Diagnostic bands	none (rails = fault)	< 0.25 V open/short-to-ground, > 4.75 V short-to-supply
Connector / pinout	Deutsch DT04-3P, A=5V B=GND C=SIG	unchanged
Harness	72-2210-3	unchanged

Effectivity

Serial number break: bodies **ASL2-26-04100** and later ship with 55-1180-1. Field replacements of a failed 55-1180-0 after stock runs out will also use 55-1180-1 and require the updated controller software. Mixed fleets are expected.

Affected variants

All ASL-200 variants (asl200_diesel_autocar, asl200_cng_peterbilt, asl200_electric_mack, asl200_diesel_mack_longreach). The reach sensor (55-1192-0 / 55-1193-0) is **not** affected.

Disposition

Item	Disposition
55-1180-0 stock on hand	Use as-is until exhausted, before serial break
WIP bodies	Complete with 55-1180-0
Service stock	Stock 55-1180-1 + software update; do not mix sensor and software revisions

Required software change

Body controller software must support the 0.5–4.5 V sensor from the serial break: sensor scaling, open/short fault thresholds, CAN database documentation and any affected diagnostics. Software released for the break must be identified in the release notes.

Approvals

Role	Name	Signature	Date
Electrical engineering	R. Mendes		
Controls software			
Manufacturing engineering	T. Lindqvist		
Quality	A. Brandt		
Service / aftermarket	K. Osei		